PROJECT REPORT STRUCTURE (STUDENT GUIDE)

Lab 4 AVCV 2026/2027: Fire-risk classification from satellite images

1. INTRODUCTION

- **Context and objective.** Introduce the task: predict one of seven FireRisk classes from an RGB satellite image. Explain the motivation and main challenges, such as class imbalance and visually similar categories. This project concerns image classification, rather than fire detection or forecasting.
- **Dataset.** Describe the supplied FireRisk subset: 5,861 images of 320 x 320 pixels, split into 3,516 training, 1,172 validation and 1,173 test images. Summarise the labelled class distribution and distinguish the original image size from the model input size.
- **Scope of the work.** Briefly introduce your CUSTOM and FINE-TUNING approaches and the main improvements or experimental questions investigated.

2. METHODOLOGY

2.1. PREPROCESSING AND DATA AUGMENTATION

- **Pipeline.** Describe resizing or cropping, RGB tensor conversion, normalisation and training-time augmentation. Explain which transformations are used for validation and test inference.
- **Justification.** Explain why the chosen augmentations are appropriate for satellite imagery. Use validation evidence to identify useful transformations and approaches that did not improve results.

2.2. SYSTEM DESIGN: CUSTOM NETWORK

- **Architecture and category rules.** Describe the CNN you designed and trained from scratch, including convolutional blocks, channels, kernel sizes, pooling, regularisation and the seven-output classifier. Show its complete architecture in the notebook, using primitive layers as required; externally supplied architectures and pretrained models are excluded from this category.
- **Design decisions.** Justify the main architectural choices and explain how your final design differs from the reference configuration. A compact diagram or layer table may support the explanation.

2.3. SYSTEM DESIGN: FINE-TUNING NETWORK

- **Pretrained model.** Identify the architecture, source of pretrained weights and pretraining dataset. Describe the replacement of the original classifier with seven outputs and any other changes. AlexNet is the provided starting point; other pretrained architectures are permitted.
- **Adaptation strategy.** State which layers were frozen or updated, how preprocessing matches the pretrained weights, and whether different parameter groups used different learning rates. Justify the selected configuration using validation results.

2.4. TRAINING PROTOCOL AND MODEL SELECTION

- **Configuration.** For both models, report the loss and any class weights, optimiser, learning rate and schedule, batch size, epoch budget, random seed and computing environment. Disclose any training-data subsampling or filtering and explain its purpose.
- **Selection and reproducibility.** Explain checkpoint selection using validation macro F1 and state whether early stopping was actually implemented. Develop and select models with training and validation data. Provide enough detail for the submitted notebook to reproduce the selected systems.

3. RESULTS AND ERROR ANALYSIS

- **Evaluation metrics.** Use macro F1 as the main classification metric, averaging the per-class F1 scores over all seven labels, 0 to 6. Supplement it with per-class precision, recall and F1, and accuracy where useful. Clearly distinguish training, validation and public CodaBench results.
- **Experimental comparison.** Summarise experiments in a concise table: category, change, objective, training budget and validation macro F1. Include the reference and selected models. Compare under similar conditions; explain differences in training data, compute or number of epochs.
- **Learning curves and confusion matrices.** Show the most informative training/validation loss and macro-F1 curves. Include validation confusion matrices with true classes in rows and predicted classes in columns. State whether values are counts or row-normalised percentages, and label all seven classes.
- **Visual error analysis.** Show representative validation images with their true and predicted labels. Identify recurring class confusions and support explanations with evidence. Discuss improvements, remaining failures and the trade-offs between CUSTOM and FINE-TUNING.
- **Competition feedback.** Report the public CUSTOM and FINE-TUNING scores and identify the corresponding submission. The leaderboard uses a fixed, undisclosed subset. Instructors retain full-test and separate private-subset results for final assessment. Public feedback alone does not determine the final performance grade.

4. CONCLUSIONS

Explain which changes had the largest supported effect on each model and what the comparison reveals about training from scratch versus fine-tuning. Reflect on the main difficulties, limitations and realistic next steps. Base conclusions on the experiments presented.

REFERENCES

Cite the FireRisk dataset paper, pretrained models and external methods or code used. Use consistent in-text citations and a corresponding reference list.

FORMAT AND INDIVIDUAL CONTRIBUTIONS

Length: three pages describing the proposed solutions, plus one additional page for tables, figures and references. Focus on experimental objectives, decisions and findings. Use numbered sections, legible figures and informative captions. Identify the group members and clearly state each member's individual improvements or experiments.

DELIVERY CHECKLIST

CodaBench: upload one prediction ZIP containing the following two files **directly at the ZIP root**, without an enclosing folder:

output_custom.csv
output_ft.csv

Each CSV must contain **all 1,173 test images x 7 probability columns**, with no header or ID column. Preserve the row order of test_reduced.csv. Use the fixed column order:

0 Water; 1 Very Low; 2 Low; 3 Moderate;
4 High; 5 Very High; 6 Non-burnable.

Aula Global: upload one group ZIP containing the report and the notebook implementing both models. Ensure that the submitted code and inference settings correspond to the submitted predictions. Follow the official assignment and Aula Global announcements for deadlines and grading rules.